

Open Data Index for Schools: Technical Report

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Introduction

The Open Data Index for Schools aims to consolidate data on the spectrum of barriers within communities (or “neighborhoods”) around the ~23,000 public (including charter) high schools in the United States. The Index reflects the level of “stress,” indicated in five domains, that school communities experience. Potential audiences for the tool are varied, as it can assist any work centered around community impact. Some potential audiences include educators, community-impact organizations, policymakers, researchers, and other community stakeholders.

The index comprises five *domains*: Economic, Education, Health, Housing, and Crime. A higher stress indicator on a domain indicates more stressful community conditions.

$$ODIS \text{ Stress Indicator} = \frac{1}{5}(\text{Economic stress}) + \frac{1}{5}(\text{Education stress}) + \frac{1}{5}(\text{Health stress}) + \frac{1}{5}(\text{Housing stress}) + \frac{1}{5}(\text{Crime stress})$$

Each domain comprises several *indicators*, each with an associated *measure*. Measures are available at different geographic levels, obtained from various sources with different updating frequencies (and different most-recent year, as many sources have several years lag in reporting) and with different geographic coverage (details on how the measures were selected are in the Annex; special attention needs to be paid to Puerto Rico, with limited coverage). See Table 1 for characteristics of index measures.

Table 1. Characteristics of Index Measures.

Domain	#	Indicator	Measure	Source	Frequency	Level	Puerto Rico	Limited extent
Economic	1	Unemployment	% 16+ unemployed but seeking work	Bureau of Labor Statistics	Annually	County	Yes	
	2	Poverty	% 0–17 in poverty	U.S. Census Bureau	Annually	Census tract	Yes	
	3	Access to broadband internet	% households with subscription to any broadband service	U.S. Census Bureau	Annually	Census tract	Yes	
	4	Single-parent household	% 0–17 living with one parent	County Health Rankings	Annually	County	No	
Education	5	Education less than high school	% 25+ less than high school	U.S. Census Bureau	Annually	Census tract	Yes	
	6	Education 2 years of college or more	% 25+ 2-year college degree or higher	U.S. Census Bureau	Annually	Census tract	Yes	
	7	Linguistic isolation	% limited-English-speaking households	U.S. Census Bureau	Annually	Census tract	Yes	
Health	8	Access to healthcare	% 6–18 with health-insurance coverage	U.S. Census Bureau	Annually	Census tract	Yes	
	9	Infant mortality rate	Infant deaths / 1,000 live births	County Health Rankings	Annually	County	No	
	10	SNAP recipients	% households with children under 18 receiving SNAP	County Health Rankings	Annually	County	Yes	
	11	Low birth weight	% live births <2.5 kg	County Health Rankings	Annually	County	No	
	12	Lead exposure risk	Lead exposure risk index	City Health Dashboard	Annually	Census tract	No	Yes*
Housing	13	Housing-vacancy rate	% housing units vacant	U.S. Census Bureau	Annually	Census tract	Yes	
	14	Housing affordability	% households spending $\geq 30\%$ income on housing	U.S. Census Bureau	Annually	Census tract	Yes	
	15	Park access	% within 10-minute walk of green space	City Health Dashboard	Static	Census tract	No	Yes*
Crime	16	Violent crime	Violent crimes /100,000	County Health Rankings	Annually	County	No	
	17	Incarceration rate	15–64 jail incarceration / 100,000	Vera Institute of Justice	Annually	County	No	

*Covers the 970 largest cities.

Domains, Indicators, and Measures

Economic

Unemployment

Description

% 16+ unemployed but seeking work

Unemployment is associated with low income, poor healthcare access, and psychological stress. It has many negative effects on physical and mental health. In general, those who are unemployed report more days of poor mental or physical health than those who are employed. Unemployment can lead to reduced self-worth manifesting as depression, anxiety, poor self-esteem, and unhealthy coping behaviors, such as alcohol and substance use. A measure, *percent unemployment rate for adults in the labor force*, is available from the Bureau of Labor Statistics. Data are updated annually and are available nationally at the county level; most recent data are for 2024.¹

Frequency	Level
Annually	County

Data source

U.S. Bureau of Labor Statistics:

bls.gov/lau/tables.htm#cntyaa

Details on how unemployment is measured at:

bls.gov/cps/eetech_methods.pdf

Additional calculations

Raw data are as percentages and are converted to rates.

Missingness and geographic extent

1.9% of counties are missing this measure (2024). No schools are missing this measure. Includes data on Puerto Rico.

¹ In some instances, data are updated annually but there are reporting lags of one or more years. County Health Rankings and City Health Dashboard use tertiary data sources and, in some cases, querying for the latest year available does not guarantee that measures will be for those years. To maintain comparability among ODIS updates, if a data source changes the reference data it uses, we use the most recent data consistent with the source in the first ODIS update. See notes about measure dates in [Data Assembly](#).

Poverty

Description

% 0–17 in poverty

Childhood poverty is associated with poor access to services, psychological stress, and poor school performance. A measure, percent children 0–17 in poverty, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district level; most recent data are for 2023.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

data.census.gov/cedsci/table?q=B17020_004E,B17020_005E,B17020_002E&tid=ACSDT5Y2023.B17020

[B17020_004E](#): 6 to 11 years

[B17020_005E](#): 12 to 17 years

[B17020_002E](#): Total

Additional calculations

$$\text{Poverty} = (\text{B17020_004E} + \text{B17020_005E}) / \text{B17020_002E}$$

Missingness and geographic extent

2.0% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure.² Includes data on Puerto Rico.

² See [notes on missingness](#).

Access to broadband internet

Description

% households with subscription to any broadband service

Access to broadband internet is increasingly important to taking advantage of education technology, for social connection, and for employment in growth industries. A measure, percent households with subscription to any broadband service, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

<http://api.census.gov/data/2023/acs/acs5/groups/B28002.html>

[B28002_004E](#): Total households with an internet subscription (broadband of any kind)

[B28002_001E](#): Total number of households

Additional calculations

Access to broadband internet = [B28002_004E](#) / [B28002_001E](#)

Missingness and geographic extent

1.8% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Single-parent household

Description

% 0–17 living with one parent

Raising a child or children as a single parent can be stress-inducing, without another parent to balance work and home life. Single-parent households may also be associated with other causes or indicators of stress.

Frequency	Level
Annually	County

Data source

County Health Rankings:

countyhealthrankings.org/sites/default/files/media/document/2023%20Data%20Dictionary%20%28PDF%29.pdf

Additional calculations

Raw data are as percentages and are converted to rates.

Missingness and geographic extent

2.7% of counties are missing this measure (2023). 1.8% of schools are missing this measure. Does not include data on Puerto Rico.

Education

Education less than high school

Description

% 25+ less than high school

Adults having a high-school diploma or equivalent are likely better able to provide for children and serve as education role models. A set of measures is available from the US Census Bureau, American Community Survey.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

<http://api.census.gov/data/2023/acs/acs5/groups/B15002.html>

[B15002_003E](#): Male high school no schooling completed (includes equivalency) 25 and over

[B15002_004E](#): Male nursery to 4th grade education 25 and over

[B15002_005E](#): Male 5th and 6th grade education 25 and over

[B15002_006E](#): Male 7th and 8th grade education 25 and over

[B15002_007E](#): Male 9th grade education 25 and over

[B15002_008E](#): Male 10th grade education 25 and over

[B15002_009E](#): Male 11th grade education 25 and over

[B15002_010E](#): Male 12th grade education, no diploma 25 and over

[B15002_020E](#): Female high school no schooling completed (includes equivalency) 25 and over

[B15002_021E](#): Female nursery to 4th grade education 25 and over

[B15002_022E](#): Female 5th and 6th grade education 25 and over

[B15002_023E](#): Female 7th and 8th grade education 25 and over

[B15002_024E](#): Female 9th grade education 25 and over

[B15002_025E](#): Female 10th grade education 25 and over

[B15002_026E](#): Female 11th grade education 25 and over

[B15002_027E](#): Female 12th grade education, no diploma 25 and over

Additional calculations

$$\text{Less than high-school} = \frac{\sum_{x \in \Omega} x}{B15002_{001E}}$$

where $\Omega =$

{B12002_{003E}, B12002_{004E}, B15002_{005E}, B15002_{006E}, B15002_{007E}, B15002_{008E}, B15002_{009E}, B15002_{010E},
{B15002_{020E}, B15002_{021E}, B15002_{022E}, B15002_{023E}, B15002_{024E}, B15002_{025E}, B15002_{026E}, B15002_{027E}}

Missingness and geographic extent

1.5% of census tracts are missing this measure. (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Education 2-year college degree or higher

Description

% 25+ 2-year college degree or higher

Adults having a college degree are likely better able to provide for children and serve as education role models. A set of measures is available from the US Census Bureau, American Community Survey.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):
api.census.gov/data/2023/acs/acs5/groups/B15002.html

[B15002_001E: Total population 25 and over](#)

[B15002_014E: Male associate's degree](#)

[B15002_031E: Female associate's degree](#)

[B15002_015E: Male bachelor's degree](#)

[B15002_032E: Female bachelor's degree](#)

[B15002_016E: Male master's degree](#)

[B15002_017E: Male professional school degree](#)

[B15002_018E: Male doctorate degree](#)

[B15002_033E: Female master's degree](#)

[B15002_034E: Female professional school degree](#)

[B15002_035E: Female doctorate degree](#)

Additional calculations

2-year college degree or higher =

$$\frac{(B15002_014E + B15002_031E + B15002_015E + B15002_032E + B15002_016E + B15002_017E + B15002_018E + B15002_033E + B15002_034E + B15002_035E)}{(B15002_001E)}$$

Missingness and geographic extent

1.5% of census tracts are missing this measure. (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Linguistic isolation

Description

% limited English-speaking households

Difficulty communicating in or understanding English limits opportunities in most communities in the US. A measure, percent limited English-speaking households, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

api.census.gov/data/2023/acs/acs5/subject/groups/S1602.html

[S1602_C03_001E](#): Limited English-speaking households

[S1602_C01_001E](#): All households

Additional calculations

Linguistic isolation = [S1602_C03_001E](#) / [S1602_C01_001E](#)

Missingness and geographic extent

1.8% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Notes

A “limited English-speaking household” is one in which no member 14 years old or over (1) speaks only English or (2) speaks a non-English language and speaks English “very well.” In other words, all members 14 years old and over have at least some difficulty with English. By definition, English-only households cannot belong to this group. Previous Census Bureau data products referred to these households as “linguistically isolated” and “Household where no one age 14 and over speaks English only or speaks English ‘very well.’”

See more in: census.gov/topics/population/language-use/about/faqs.html

Health

Access to healthcare

Description

% 6–18 with health-insurance coverage

Access to healthcare is important to maintaining good health, and poor access is stressful to a community. Measuring access is difficult, as low healthcare usage could indicate good health and so low demand, or poor health and low supply or inability to pay. A measure, percent children 6–18 with health-insurance coverage, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023. And a similar measure, percent children 6–18 with public health-insurance coverage, is also available.

Frequency	Level
Annually	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

api.census.gov/data/2023/acs/acs1/subject/groups/S2701.html

[S2701_C03_003E](#): Percent of civil noninstitutionalized population 6–18 that has health insurance.

Missingness and geographic extent

2.2% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Infant mortality rate

Description

Number of all infant deaths (within 1 year) per 1,000 live births

Morbidity and early mortality are strong indicators of stress. An especially telling indicator is infant mortality, which is rare but notably high in especially stressed communities. A measure, infant deaths (within 1 year) per 1,000 live births, is available from County Health Rankings, derived from data from the National Center for Health Statistics, National Vital Statistics System. Data are updated annually and are available nationally, at the county level; most recent data are for 2023.

From County Health Rankings: “Infant mortality represents the health of the most vulnerable age group and can help with interpreting the years of potential life lost (YPLL) rate in a county. Infant mortality is also commonly used to examine global health differences, as well as to understand historic racial inequities in the US.”

Frequency	Level
Annually	County

Data source

County Health Rankings:

countyhealthrankings.org/sites/default/files/media/document/2023%20Data%20Dictionary%20%28PDF%29.pdf

Missingness and geographic extent

61.4% of counties are missing this measure (2023). 24.7% of schools are missing this measure. Most of the missingness is likely attributable to low infant-death rates. See note below. No data on Puerto Rico.

Notes

“Infant death is a relatively rare event in most counties.”

“A missing value is reported for counties with fewer than 20 infant deaths in the time frame.”

SNAP recipients

Description

% households with children under 18 receiving SNAP

SNAP is an important program for supporting children in low-income households, to ensure that they receive adequate nutrition, critical to good health; a high incidence of SNAP recipients indicates challenges to nutrition in the community. A measure, percent households with children under 18 receiving SNAP, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023.

Frequency	Level
Annual	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

api.census.gov/data/2023/acs/acs5/subject/groups/S2201.html

[S2201_C04_009E](#): Estimate. Percent households receiving food stamps/SNAP. Households with children under 18 years.

Additional calculations

Raw data are as percentages and are converted to rates.

Missingness and geographic extent

5.3% of census tracts are missing this measure (2023). 1.3% of schools are missing this measure. Includes data on Puerto Rico.

Low birth weight

Description

% live births < 2.5 kg

Babies born with low birth weight are more likely to experience long-term physical-health problems and cognitive-development challenges later in life. Low birth weight is associated with poor healthcare access and maternal behavioral-health problems. There are significant demographic differences in low-birth-weight prevalence, with babies born to Black mothers more likely to be low birth weight compared to babies born to White, non-Hispanic mothers.

A measure, percent of live births with birth weight < 2.5 kg, is available from County Health Rankings, derived from data from the National Center for Health Statistics, National Vital Statistics System.

Frequency	Level
Annual	County

Data source

County Health Rankings:

countyhealthrankings.org/sites/default/files/media/document/2023%20Data%20Dictionary%20%28PDF%29.pdf

Additional calculations

Raw data are as percentages and are converted to rates.

Missingness and geographic extent

5.9% of counties are missing this measure (2023). 2.2% of schools are missing this measure. No data on Puerto Rico.

Notes

From County Health Rankings: “Births are counted in the county corresponding to the mother’s address on the child’s birth certificate, not the county the child was born in.”

Lead exposure risk

Description

Lead exposure risk index (see notes below)

Exposure to lead can seriously damage a child’s brain and nervous system, impair growth and development, and yield learning or behavioral problems and hearing/speech damage. Living in high-lead-risk census tracts is associated with greater harms for children from lower- versus higher-income families—lower cognitive test scores, smaller cortical surface area, and smaller cortical volume.

Frequency	Level
Annual	Census tract

Data source

City Health Dashboard:

cityhealthdashboard.com/metric/lead-exposure-risk-index

Most recent data from 2022.

More at cityhealthdashboard.com/technical-documentation

Missingness and geographic extent

This measure is available only for the five hundred most-populous cities in the United States. 55.8% of census tracts are missing this measure. 52.1% of schools are missing this measure. No data on Puerto Rico.

Notes

From the data source:

Housing with potential lead risk

We count the number of housing units in each of five time periods: pre-1939, 1940–59, 1960–79, 1980–99, and 2000 or newer. The count of housing units in each time period is weighted by the likelihood of lead exposure in housing of that era, which results in an overall percent of area housing likely to have some risk of lead exposure.

Lead exposure risk index

We took the housing with potential lead risk and factored in information about the percentage of households living at or below 125% of the poverty level. We standardized, weighted, summed, and ranked these values from 1, or lowest risk, to 10, or highest risk, to create a scale of overall lead exposure risk.

Housing

Housing–vacancy rate

Description

Vacant housing may be an indicator of neighborhood stress—declining population, high foreclosures, or overbuilding preceding an economic crash. A measure, percent housing units vacant, is available from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023.

Frequency	Level
Annual	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):
api.census.gov/data/2023/acs/acs1/profile/groups/DP04.html
[DP04_0003E](#): vacant housing units
[DP04_0001E](#): total housing units

Additional calculations

[DP04_0003E](#) / [DP04_0001E](#)

Missingness and geographic extent

1.8% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Housing affordability

Description

Share of households spending more than 30% of income on housing

While high absolute housing costs often reflect wealthy, low-stress neighborhoods, lower-income neighborhoods in high-cost regions can be beset with high housing-cost burdens. A measure, percent households that spend 30% or more of their income on housing, can be derived from data from the US Census Bureau, American Community Survey. Data are updated annually and are available nationally, at the tract, ZCTA, and school-district levels; most recent data are for 2023.

Frequency	Level
Annual	Census tract

Data source

American Community Survey 5-year estimates (ACS-5):

api.census.gov/data/2023/acs/acs5/subject/groups/S2503.html

The number of housing units by household income bracket is available. The percentage of housing units that spend 30% or more of household income in housing is also available by income bracket. Using these variables, it is possible to calculate the share of households spending 30% or more of income on housing.

$$\frac{\sum_{i \text{ income brackets}} x_i \cdot y_i}{n},$$

where x_i is the number of occupied households in bracket i , y_i is the percentage of occupied households in bracket i that spend 30% or more of their income on housing, and n is the total number of occupied units.

Since the predefined brackets in x and y are not equal (see variables below), we grouped them so they are comparable.

Additional calculations

Owner-occupied housing units³

$$\frac{(002E + 003E + 004E + 005E).028E + 007E.032E + (008E.036E) + 009E.040E + (010E + 011E + 012E) * 044E}{001E.001E}$$

³ Variable names are shortened—all have the S2503_CO1_ prefix.

Where:

[S2503_C01_001E](#): Occupied housing units (estimated).

[S2503_C02_002E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) less than \$5,000.

[S2503_C02_003E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$5,000 to \$9,999.

[S2503_C02_004E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$10,000 to \$14,999.

[S2503_C02_005E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$15,000 to \$19,999.

[S2503_C02_028E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income of less than \$20,000.

[S2503_C02_006E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$20,000 to \$24,999.

[S2503_C02_007E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$25,000 to \$34,999.

[S2503_C02_032E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income \$20,000 to \$34,999.

[S2503_C02_008E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$35,000 to \$49,999.

[S2503_C02_036E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income \$20,000 to \$49,999.

[S2503_C02_009E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$50,000 to \$74,999.

[S2503_C02_040E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income \$50,000 to \$74,999.

[S2503_C02_010E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) 75,000 to \$99,999.

[S2503_C02_011E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) 100,000 to \$149,999.

[S2503_C02_012E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) 150,000 or more.

[S2503_C02_044E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income \$75,000 or more.

Missingness and geographic extent

1.8% of census tracts are missing this measure (2023). 1.2% of schools are missing this measure. Includes data on Puerto Rico.

Park access

Description

% population living within a 10-minute walk of green space

Public parks are critical resources for physical activity in underserved communities. Even mental health is strongly associated with residential distance from parks. Parks can also have a positive impact on environmental outcomes like air pollution. A measure, percent population living within a 10-minute walk of a green space, from ParkServe, was published in 2018, available nationally at the tract level. These data are also available to us through a data repository maintained by NYU Population Health.

From City Health Dashboard: “Access to parks has been associated with better mental and physical health, lower death rates, and improved social connection. While a growing body of research is making the evidence base linking park access and health stronger, the reasons for this link are not yet fully clear. Research has shown that green space may be associated with increased physical activity, reduced stress, improved quality of life, and better mental health across different groups of people. Also, parks and green space may be associated with increased neighborhood social ties by serving as meeting places for local residents and spaces for community events. Parks may also have positive impacts on environmental outcomes such as reducing air pollution and extreme heat, especially in urban areas.”

Estimates for this measure are from 2024 ParkServe data. The City Health Dashboard sources ParkServe data from 2023 or 2024 (it is unclear from their documentation); ParkServe is using demographic data from 2023 or 2024 and geographic data from 2010, likely because that information changes less frequently. We do not find the update frequency for demographic and geographic data on the ParkServe website.

Frequency	Level
	Census tract

Data source

City Health Dashboard cityhealthdashboard.com/metric/park-access
More on cityhealthdashboard.com/technical-documentation
and tpl.org/parkserve/about

Missingness and geographic extent

This measure is available only for the five hundred most-populous cities in the United States. 54.2% of census tracts are missing this measure. 51.9% of schools are missing this measure. No data on Puerto Rico.

Crime

Violent crime

Description

Number of reported violent-crime offenses per 100,000 population

Violent crime exacts a toll on victims and the community.⁴ Crime data are notoriously fickle, as jurisdictions vary in their definitions of crime and reporting by local law-enforcement agencies to the FBI and BJS is unreliable. Murder is the most difficult crime to sweep under the rug; while rare, it is generally correlated with other violent crime in larger jurisdictions and so is a good proxy. In smaller jurisdictions, which may have zero or one murders a year, it is a very noisy indicator. Also, while actual violent-crime rates and public perceptions can change very quickly, reporting and data collection are slow so that “uniform,” national data are typically reported with a lag.

A measure, number of reported violent-crime offenses per 100,000 population, was available from County Health Rankings, sourced from the FBI Uniform Crime Reporting Program. Data were updated annually until 2022 and were available nationally at the county level.⁵

Frequency	Level
Annual (lagged)	County

Data source

County Health Rankings:

web.archive.org/web/20221008215511/https://www.countyhealthrankings.org/explore-health-rankings/measures-data-sources/county-health-rankings-model/health-factors/social-and-economic-factors/community-safety/violent-crime-rate

Additional calculations

All observations with a violent-crime rate value higher than the 95th percentile of the variable were assigned that value instead.

⁴ Violent crimes are defined as offenses that involve face-to-face confrontation between a victim and a perpetrator, including homicide, rape, robbery, and aggravated assault.

⁵ County Health Rankings has new measures of community safety, at [countyhealthrankings.org/health-data/health-factors/social-economic-factors/community-safety](https://www.countyhealthrankings.org/health-data/health-factors/social-economic-factors/community-safety). The FBI UCR continues to collect this measure, but it is not readily acquired at the county level. Future ODIS updates will revisit this measure.

Missingness and geographic extent

58% of counties are missing this measure (2022). 24.9% of schools are missing this measure. No data on Puerto Rico.

Incarceration rate

Description

15–64 jail incarceration / 100,000

Incarceration imposes stress on the communities that incarcerated people leave behind. Prison population figures are generally complete and timely, but there is no uniform collection of data on where prison inmates are from. Jails mostly house people who are from the jurisdiction that they are in (typically counties); while jail-population data have been slow to be collected and reported, starting in 2024 BJS is conducting an annual census. A measure, jail incarceration rate per 100,000 residents 15–64, is available from the Vera Institute of Justice. This is a new mechanism; data will be updated periodically and are available nationally at the county level. Most recent data are for 2023.

Frequency	Level
Annual	County

Data source

Vera Institute of Justice:
github.com/vera-institute/incarceration-trends

Missingness and geographic extent

39.9% of counties are missing this measure (2023). 23% of schools are missing this measure. No data on Puerto Rico.

Index Calculation

Notes on pre-processing

To construct a composite index, all measures must be expressed in similar fashions (“commensurate”). As the raw data associated with the measures for this index are expressed in a variety of fashions (percentages, rates, etc.), they must be transformed (“pre-processed”) to be expressed similarly—as numbers from 0 to 100.

The raw data associated with the measures selected for the index do not all have the same sign (or “directionality”); i.e., “more” of one measure may indicate better conditions in the community, whereas more of another may indicate worse conditions. For example, more *unemployment* in a given census tract indicates higher community stress, whereas more *access to health-insurance coverage* indicates lower community stress. Measures for which the raw data have lower values associated with higher stress are transformed by reversing the sign.

Different measures have different ranges of actual values (on the 0–100 scale). To capture the full extent of variation by community, for each measure, some measures are truncated, so that all values above or below selected “cut points” are treated as, respectively, being the minimum or maximum value of that measure. For example, when the poverty measure is greater than 40%, the raw data are truncated to 40%, which becomes a score of 100.

Table 2 summarizes the pre-processing for each measure.

Table 2. Measures Pre-Processing.

Domain	#	Indicator	Measure	Original scale	Scaled range	Updated cuts
Economic	1	Unemployment	% 16+ unemployed but seeking work	%	+	
	2	Poverty	% 0–17 in poverty	%	+	>0.4 → 0.4
	3	Access to broadband internet	% households with subscription to any broadband service	%	–	<0.4 → 0.4
	4	Single-parent household	% 0–17 living with one parent	%	+	
Education	5	Education less than high school	% 25+ less than high school	%	+	
	6	Education 2 years of college or more	% 25+ 2-year college degree or higher	%	–	
	7	Linguistic isolation	% limited English-speaking households	%	+	>0.25 → 0.25
Health	8	Access to healthcare	% 6–18 with health-insurance coverage	%	–	
	9	Infant mortality rate	Infant deaths / 1,000 live births	≥0	+	
	10	SNAP recipients	% households with children under 18 receiving SNAP	%	+	
	11	Low birth weight	% live births <2.5 kg	%	+	
	12	Lead exposure risk	Lead exposure risk index	1–10	+	
Housing	13	Housing-vacancy rate	% housing units vacant	%	+	>0.4 → 0.4
	14	Housing affordability	% households spend ≥50% income on housing	%	+	
	15	Park access	% within 10-minute walk of green space	%	–	
Crime	16	Violent crime	Violent crimes /100,000	≥0	+	95 th percentile
	17	Incarceration rate	15–64 jail incarceration / 100,000	≥0	+	95 th percentile

Data assembly: School Attendance Boundary (SAB)

The intent of the index is to characterize the challenges faced by public high schools and their student bodies, in order to leverage resources and put education outcomes data into context. The natural unit of analysis for the index is *a public high school*. For this purpose, ideally, we would have data reflecting each measure on the well-defined geographic area from which the school's students are drawn. Many schools have such an area, which the National Center for Education Statistics, NCES, defines as a School Attendance Boundary (SAB);⁶ some schools, such as magnet schools or those in districts with open enrollment (as in New York City), draw from an entire school district. And some schools do not have an NCES SAB, for reasons unspecified.

Unfortunately, none of the measures for this index are collected at the SAB level. So, we *synthesize* measures at the SAB level, from data collected at other levels. The smallest geographic units for which there are publicly available data on the measures are census tract or county, depending on the measure (see Table 1). We conduct an “assembly process,” where data at the census-tract or county level are assigned to schools associated with the geographic units and using SABs, when possible.

The first step in the assembly process is the creation of two data files (in .csv format) that contain all measures collected at the county level and at the census-tract level, separately. This is done by downloading or collecting information from the data sources detailed above, using API calls or https requests and attaching the information to reference tables on the universe of counties and census tracts (see Data sources, below).

Once these data are available, measure values are assigned to each school depending on the following logic, which applies for measures that are available at the census-tract level:

1. If an SAB is available (n~12,807), then all intersecting census tracts⁷ are selected, and their measure values population-weighted and averaged.
2. If an SAB is not available (n~10,792), the school's location (geographic coordinates) is used to determine the census tract in which it is located. ZCTAs are reasonable proxies for school “neighborhoods” and correspondence tables between ZCTAs and census tracts are available.⁸ Measures from the census tracts that intersect the school's zip code are population weighted and averaged.

All schools in New York City have the same SAB, the geographic extent of the city, due to the school district's open-enrollment policy. The strategy in #2 is applied in this case, as well as for four schools with open enrollment in the Los Angeles Unified School District.

For measures available only at the county level, all schools in the county have the same value.

⁶ An SAB has an associated shapefile, which defines the geographic boundaries.

⁷ Census tracts and counties also have shapefiles, so it can be determined which census tracts and counties intersect with which SABs. Almost always, an SAB will lie entirely within one county and will intersect with multiple census tracts.

⁸ See [census.gov/geographies/reference-files/2010/geo/relationship-files.html#par_textimage_674173622](https://www.census.gov/geographies/reference-files/2010/geo/relationship-files.html#par_textimage_674173622).

With all schools having values assigned for each measure and each domain, the composite index values can now be calculated:

$$\text{Index} = \sum_{i \text{ in domains}} \sum_{j \text{ in metrics in domain } i} d_i m_{ij} x_{ij},$$

where d_i is the weight of domain i relative to the value of the index; m_{ij} is the weight of measure j in domain i , relative to the domain; and x_{ij} is the standardized value of measure i in domain j .

In simpler terms, for each school, we create a score for each domain, which is a weighted average of the measures in that domain. If a domain has, say, three indicators, x , y , and z , each with a possible range of 0–100, then the score for that domain is $a*x + b*y + c*z$, where the weights a , b , and c sum to one. The weights indicate how much relative importance we put on each of the measures—they are subjective choices. For simplicity, we make each of the weights in a domain equal (in the example above, the weights a , b , and c are all $1/3$).⁹

The index is then a weighted sum of the five domain scores. Again, we make the weights equal, so each is $1/5$.

Notes on missingness

If a measure in a domain is missing for a school, the weights of all other measures in the domain are re-calculated so that their sum is 1.0, proportionally to the original intended weights. For instance, if four measures in a domain had weights [0.5, 0.15, 0.2, 0.15] and the first measure is missing, then the remaining measures, which sum to 0.5, are adjusted to [0.0, 0.3, 0.4, 0.3].

If all measures within a domain are missing, then the entire domain is marked as N/A. Similarly, if one of the domains is missing, then the weights of the remaining domains are adjusted proportionally to their original weights.

For measures available at the census tract and schools with known SAB, a school will be marked as missing such measure if and only if *all* of the census tracts intersecting its SAB are missing it. For measures available at the county level, if a measure is missing in a county, all schools within that county will be marked as missing it. For measures available at the census-tract level and schools with no known SAB, if a measure is missing in a census tract, all schools within that census tract will be marked as missing it.

⁹ The weights are adjustable in the developer tool.

Data sources

Three kinds of information were needed to build the index:

- Information about schools, since schools are the unit of analysis of the index.
 - School-attendance boundaries (SABs) for 14,172 high schools. See nces.ed.gov/programs/edge/sabs. Data are from the [2015–2016 NCES School Attendance Boundary Survey \(SABS\)](#).
 - [List of high schools for the 2024–2025 school year](#) from the NCES Common Core of Data, CCD. Includes information on each school, such as name, geographic coordinates, zip code, county, city, and state, for 23,790 high schools. This is our universe of schools.
- Information about geographic units.
 - Census-tract populations were gathered to implement the weighting scheme in strategy #2, when needed. This information comes from the US Census Bureau.
 - [Geographic boundaries of census tracts](#). These are used to determine which schools belong to which census tracts, which is necessary to strategy #2.
 - [2010 ZCTA to Census Tract Relationship File Layout](#). [This list](#) matches ZCTAs with all census tracts where there is a non-null geographic intersection. This is needed for when strategy #2 in the assembly process is implemented.
 - [Total county populations](#) were added to the data set for consistency.
- Source data for measures: See below

Additionally, NAF provided the list of schools they partner with. It was added to the database and is an optional column in the online index display.

For measures:

From census.gov/data/developers/data-sets/acs-5year.html. There are different API call structures for each kind of table:

- **Detailed Tables** contain the most detailed cross-tabulations, many of which are published down to block groups. The data are population counts. There are over 20,000 variables in this dataset.
 - [B28002_004E](#): Households with an internet subscription of any kind
 - [B28002_001E](#): Presence and types of internet subscriptions in household
 - [B15002_001E](#): Total population—sex by educational attainment for the population 25 years and over
 - [B15002_011E](#): Male—sex by educational attainment for the population 25 years and over
 - [B15002_028E](#): Female—sex by educational attainment for the population 25 years and over
 - [B17020_002E](#): Total population—income in the past 12 months below poverty level
 - [B17020_004E](#): Ages 6 to 11—income in the past 12 months below poverty level
 - [B17020_005E](#): Ages 12 to 17—income in the past 12 months below poverty level

- **Subject Tables** provide an overview of the estimates available on a particular topic. The data are presented as population counts and percentages. There are over 18,000 variables in this dataset.

ACS-5 subject tables that were used for the index:

- [S1602_C03_001E](#): Limited English-speaking households
- [S1602_C01_001E](#): Total—limited English-speaking households
- [S2701_C03_003E](#): Percent insured—civilian noninstitutionalized population—6 to 18 years of age
- [S2201_C04_009E](#): Percent households with children under 18 years receiving food stamps
- [S2503_C01_001E](#): Occupied housing units (estimated).
- [S2503_C02_002E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) less than \$5,000
- [S2503_C02_003E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$5,000 to \$9,999
- [S2503_C02_004E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$10,000 to \$14,999
- [S2503_C02_005E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$15,000 to \$19,999
- [S2503_C02_028E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income (in 2023 inflation-adjusted dollars) less than \$20,000
- [S2503_C02_006E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$20,000 to \$24,999
- [S2503_C02_007E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$25,000 to \$34,999
- [S2503_C02_032E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income (in 2023 inflation-adjusted dollars) \$20,000 to \$34,999
- [S2503_C02_008E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$35,000 to \$49,999
- [S2503_C02_036E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income (in 2023 inflation-adjusted dollars) \$20,000 to \$49,999
- [S2503_C02_009E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$50,000 to \$74,999
- [S2503_C02_040E](#): Percent occupied housing units with monthly housing costs of 30 percent of more of household income in the last 12 months—for households with annual income (in 2023 inflation-adjusted dollars) \$50,000 to \$74,999
- [S2503_C02_010E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$75,000 to \$99,999
- [S2503_C02_011E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$100,000 to \$149,999

- [S2503_C02_012E](#): Percent occupied housing units with household income in the past 12 months (in 2023 inflation-adjusted dollars) \$150,000 or more
- [S2503_C02_044E](#): Percent occupied housing units with monthly housing costs of 30 percent or more of household income in the last 12 months—for households with annual income (in 2023 inflation-adjusted dollars) \$75,000 or more.
- **Data Profiles** contain broad social, economic, housing, and demographic information. The data are presented as population counts and percentages. There are over 1,000 variables in this dataset.
ACS-5 profile tables that were used in the index:
 - [DP04_0003E](#): Vacant housing units
 - [DP04_0001E](#): Total housing units
- **Comparison Profiles** are similar to Data Profiles but also include comparisons with past-year data. The current-year data are compared with prior 5-year data and include statistical-significance testing. There are over 1,000 variables in this dataset.

In some instances, data are updated annually but there are reporting lags of one or more years. County Health Rankings and City Health Dashboard use tertiary data sources and, in some cases, querying for the latest year available does not guarantee that measures will be for those years.

Notes on demographic context data

Demographic characteristics of relevant populations are provided as contextual information in the online index tool. Note that these demographic data are *not* inputs to the Index, but they may help the user interpret the Index values. The displayed information on race and ethnicity is gathered from the American Community Survey 5-year estimates¹⁰.

About race: “[T]he U.S. Census Bureau collects race data in accordance with guidelines provided by the U.S. Office of Management and Budget (OMB), and these data are based on self-identification. The racial categories included in the census questionnaire generally reflect a social definition of race recognized in this country and not an attempt to define race biologically, anthropologically, or genetically. In addition, it is recognized that the categories of the race question include race and national origin or sociocultural groups. OMB requires that race data be collected for a minimum of five groups: White, Black or African American, American Indian or Alaska Native, Asian, and Native Hawaiian or other Pacific Islander. OMB permits the Census Bureau to also use a sixth category – Some Other Race. Respondents may report more than one race.”¹¹

About ethnicity: “[T]he U.S. Census Bureau adheres to the U.S. Office of Management and Budget's (OMB) definition of ethnicity. There are two minimum categories for ethnicity:

¹⁰ Tables [B02001](#) (race) and [B03002](#) (ethnicity).

¹¹ From [census.gov/glossary](https://www.census.gov/glossary).

Hispanic or Latino and Not Hispanic or Latino. OMB considers race and Hispanic origin to be two separate and distinct concepts. Hispanics and Latinos may be of any race.”¹²

The race and ethnicity categories provided in the tool are:

- White alone
- Black or African American alone
- American Indian and Alaska Native alone
- Asian alone
- Native Hawaiian and Other Pacific Islander alone
- Some other race alone
- Two or more races
- Hispanic or Latino

Notes on inclusion

Data from Puerto Rico are collected for nine of the sixteen measures. Those measures are unemployment, poverty, access to broadband internet, high-school graduates, linguistic isolation, access to healthcare, SNAP recipients, housing-vacancy rate, and housing affordability.

Racial and ethnic minorities (i.e., other than “non-Hispanic Whites”) are systematically undercounted in the American Community Survey data collected by the U.S. Census Bureau. While undercounted communities are disserved when associated census data are used for *population-based* apportionment of resources or representation, they are of less concern for our present purposes. That is, census tracts are, by design, small and relatively homogeneous. So, even if smaller *shares* of the population in high-minority tracts are surveyed than in low-minority tracts, the people surveyed nonetheless accurately reflect the populations in their tracts, so there is no systematic bias in the tract-level *characteristics*.

¹² From [census.gov/glossary](https://www.census.gov/glossary).